# P1-21 — Independent Proof Audit of the Proposed Consequences R1–R4

**Object audited:** P1_20_NEW_CONSEQUENCES.md and code/validate_p1_20.py, against the frozen blueprint (02_THEOREM_BLUEPRINT.md, SHA-256 ec2fd8d58f8d2a986314f83997d750d689f7c88e98ec8f87753ab4c9cf438cff, unchanged), the manuscript (Theorems 5.2, 6.1, 6.3; Corollaries 7.3, 7.4; Proposition 6.4), and the proof packages P1-04/P1-05/P1-06. Repository at audit start: commit 0da7e45, local = remote, clean.

**Audit method.** Each proof was re-derived from the definitions, not checked against its own text. Exact computation (fresh audit-side enumeration, Section 6) was used only as corroboration. **No adoption decision is made; R1–R4 and N1–N3 all remain PROPOSED. The frozen blueprint and manuscript are not edited. No novelty claim is made; Gate G1 remains DEFERRED.**

**Standing notation.** One orbit: length L >= 1, weight w >= 1, exponents alpha_0,...,alpha_{L-1} in {0,...,P}, indices mod L. Per-orbit contributions (T4/manuscript Theorem 5.2): code K(alpha) = w ∗ sum_i (P - alpha_i), hull H(alpha) = w ∗ sum_i min(P - alpha_i, alpha_{i-1}). Global dimensions are sums over orbits/components; D := sum_O P w_O L_O is the maximum code dimension (shown below to equal n ∗ sum_s m_s). Uniform labeled model for probabilistic statements. Combinatorial identities are in Z[u,z] for integer P >= 1; code realization needs the frozen hypotheses.

**Verdicts in one line each** (details and proofs below): **R1 correct** (complete involution proof; genuinely new joint content). **R2 correct** (matrices, orientation, boundary conditions verified; genuine distribution refinement of C4; one notation overload flagged). **R3 correct for every P >= 1, odd or even** (but it is the normalized form of N3 — not an independent second result). **R4 correct** (generic Cayley–Hamilton recurrence; only the P=1 case is an explicit closed form — scope must be stated that way). **No statement among R1–R4 is refuted; no correction to any statement is required.** Two wording fences are recorded in Section 7.

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## 1. R1 — the symmetry N(K,H) = N(D-K,H)

### 1.1 The map, stated on the full labeled assignment space

A global labeled assignment is the disjoint union of its orbit restrictions alpha^{(O)} in {0,...,P}^{L_O} (manuscript Theorem 6.3 / T1). Define Phi orbit by orbit: for alpha in {0,...,P}^L,

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Phi(alpha)_i := P - alpha_{-i}        (indices modulo L).
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**Phi is an involution.** Phi(Phi(alpha))_i = P - Phi(alpha)_{-i} = P - (P - alpha_{-(-i)}) = alpha_i. Hence Phi is a bijection of each orbit cube, and the orbitwise product map is a bijection (indeed an involution) of the full labeled assignment space (Cartesian product over all (s,O)). No hypothesis beyond the orbit decomposition is used; in particular compatibility/field structure is irrelevant to this combinatorial step — it enters only through T4/T5 identifying the cube sums with code/hull dimensions.

### 1.2 Code dimension transforms as claimed

K(Phi(alpha)) = w ∗ sum_i (P - Phi(alpha)_i) = w ∗ sum_i alpha_{-i} = w ∗ sum_i alpha_i (reindexing a cyclic sum). Since K(alpha) = w ∗ sum_i (P - alpha_i) = wPL - w ∗ sum_i alpha_i, we get K(Phi(alpha)) = PwL - K(alpha) per orbit. Summing over orbits: dim C(Phi(a)) = D - dim C(a) with D = sum_O P w_O L_O. (Consistency of D: per component sum_O w_O L_O = m_s ∗ sum_{f in F_s} deg f = m_s n_0, the distinct irreducible factors of x^{n_0} - mu_s having degrees summing to n_0; hence D = P n_0 sum_s m_s = n sum_s m_s, the ambient F_q-dimension — as it must be, since dim C + dim D(C) = n sum_s m_s.)

### 1.3 Hull dimension is preserved

H(Phi(alpha)) = w ∗ sum_i min(P - Phi(alpha)_i, Phi(alpha)_{i-1}) = w ∗ sum_i min(alpha_{-i}, P - alpha_{-i+1}). Substituting j = -i + 1 (a bijection of Z/LZ), the i-th term is exactly the original hull

term at position j, namely min(P - alpha_j, alpha_{j-1}). So the hull sums are equal **term by term after reindexing** — no averaging or identity of totals is needed. Therefore H(Phi(alpha)) = H(alpha) per orbit, and globally dim Hull(Phi(a)) = dim Hull(a).

Combining 1.2–1.3: Phi pairs assignments of joint type (K,H) with assignments of type (D-K,H), bijectively. Hence N(K,H) = N(D-K,H) for all integers K,H (with N = 0 off the support), equivalently E(u,z) = u^D E(u^{-1},z) as polynomials (the right side is a polynomial because E has u-degree at most D and the identity is coefficientwise). Corollary R1.1 follows: summing over H gives Pr(dim C = K) = Pr(dim C = D-K), so whenever the conditioning value has positive probability the normalized conditional laws of dim Hull at K and D-K coincide, and so do all conditional moments. ∎

### 1.4 Edge cases and hypothesis check

- L = 1: Phi(alpha_0) = P - alpha_0 (reversal is trivial); hull term min(P-a,a) is invariant under a -> P-a; code K -> Pw - K. Holds.
- P = 1: exponents binary; Phi is complement composed with reversal; hull counts 1->0 transitions, preserved by the termwise computation above. Holds.
- w > 1: w is a common factor of both contributions; the argument is unchanged (audited at w = 2, 3).
- The reversal is what makes 1.3 termwise. (Audit remark, **not** a claim: dropping the reversal — complement alone — produced no counterexample on the audited grid P <= 4, L <= 4, so this audit does **not** assert that reversal is necessary; R1's proof neither needs nor makes that assertion. No use is made of this remark.)
- Hypotheses: the identity for the ∗enumerator polynomial∗ is exactly as strong as T5/T6; no new field hypothesis is introduced. For P not a prime power it remains a true polynomial identity about the cube sums, with no code interpretation — same status as T5's remark in the manuscript.

### 1.5 Verdict and genuine content

**R1 is CORRECT, with a complete proof.** Genuine content: C2 gives only the code-dimension marginal (the H-summed shadow); the joint symmetry at fixed hull dimension and the conditional-law symmetry (R1.1) are not in C1–C7 or N1–N3. The proof is elementary (one involution), but the statement is a real structural fact about E(u,z), e.g. it halves the independent joint coefficients and forces the symmetric conditional-mean pattern seen in Section 5 of P1-20.

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## 2. R2 — restricted self-orthogonal / dual-containing enumerators

### 2.1 From the C4 criteria to the matrices

Audited inputs (manuscript Corollary 7.4, proved in P1-05 Section 5): on every orbit edge (previous a = alpha_{i-1}, current b = alpha_i, indices mod L):

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C(a) self-orthogonal  iff  a + b >= P on every edge;   then Hull = C, so dim Hull = dim C.
C(a) dual-containing  iff  a + b <= P on every edge;   then Hull = D(C).
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Also from T3/Corollary 7.4's proof: per orbit, dim D(C) = w ∗ sum_i alpha_{i-1} = w ∗ sum_i alpha_i (cyclic reindexing), while dim C = w ∗ sum_i (P - alpha_i).

**Self-orthogonal matrix.** S_w(t)_{a,b} = t^{w(P-b)} if a+b >= P, else 0. In tr(S_w(t)^L), the trace expansion (as in T5) sums over closed walks a_0 -> a_1 -> ... -> a_{L-1} -> a_0; the step a_{j-1} -> a_j contributes the factor for edge j, including the **closing edge** a_{L-1} -> a_0, which is precisely the orbit-boundary condition (edge 0 with predecessor alpha_{L-1}). Every closed walk is one exponent tuple, admissibility is enforced edge by edge, and the product of step weights is t^{w ∗ sum_j (P - a_j)} = t^{K(alpha)}. Hence tr(S_w(t)^L) = sum_{alpha SO on this orbit} t^{K(alpha)}. Setting t = u z gives the joint weight u^{K} z^{K} = u^{dim C} z^{dim Hull} on this family. **Orientation check:** the exponent uses the ∗current∗ vertex b, matching T5's convention (row =

previous, column = current); because both contributions are cyclic sums, a transposed reading would give the same trace polynomial here, but the stated orientation is the one consistent with T5. **Boundary L=1:** the trace is the diagonal sum over 2a >= P of t^{w(P-a)} — the self-loop condition alpha_0 + alpha_0 >= P, correct.

**Dual-containing matrix.** D_w(u,z)_{a,b} = u^{w(P-b)} z^{w a} if a+b <= P, else 0. A closed walk contributes u^{w sum_j (P-a_j)} ∗ z^{w sum_j a_{j-1}} = u^{K(alpha)} z^{w sum_i alpha_i}; on admissible (dual-containing) tuples, w sum_i alpha_i is the per-orbit hull dimension because Hull = D(C) and dim D(C) has the displayed form. So tr(D_w(u,z)^L) = sum_{alpha DC on this orbit} u^{K(alpha)} z^{H(alpha)}, with the closing edge enforcing the boundary condition at position 0 exactly as above; L=1 gives the diagonal sum over 2a <= P, correct.

**Global assembly.** Both admissibility conditions are conjunctions over orbits and both dimensions are sums over orbits, so the finite distributive law (as in T6) gives the global products prod_s prod_O tr(...^{L_O}). Specializing u = z = 1 makes every admissible step weight 1, recovering precisely the C4 counts prod tr(B_P^{L_O}) and prod tr((B'_P)^{L_O}). ∎

## 2.2 The symmetry S(K) = Dcal(D-K, K) (notation renamed here, see 2.4)

Let N_SO(K) = number of self-orthogonal assignments with (dim C, dim Hull) = (K,K), and N_DC(K,H) = number of dual-containing assignments with (dim C, dim Hull) = (K,H). **Claim: N_SO(K) = N_DC(D-K, K).** Proof: the per-orbit complement beta_i = P - alpha_i (no reversal) maps edge sums a+b >= P to (P-a)+(P-b) <= P, hence is a bijection from self-orthogonal to dual-containing assignments (applied orbitwise; inverse is itself). If alpha is self-orthogonal, then dim C(beta) = sum_O w_O sum_i alpha_i = D - dim C(alpha) = D - K, and dim Hull(beta) = dim D(C(beta)) = sum_O w_O sum_i beta_i = sum_O w_O sum_i (P - alpha_i) = K (equivalently D - dim C(beta)). Hence the bijection pairs the two counted sets. ∎ Summing over K recovers C4's "the two counts are equal" — the claim is its distribution-level refinement.

## 2.3 The P=2, L=2, w=1 example reproduced directly

All nine tuples, with (K,H) = (sum(2-alpha_i), sum min(2-alpha_i, alpha_{i-1})) and edge sums taken cyclically:

| alpha | (K,H) | edge sums | SO (>=2)? | DC (<=2)? |
|---|---|---|---|---|
| (0,0) | (4,0) | 0,0 | no | yes |
| (0,1) | (3,1) | 1,1 | no | yes |
| (0,2) | (2,2) | 2,2 | yes | yes |
| (1,0) | (3,1) | 1,1 | no | yes |
| (1,1) | (2,2) | 2,2 | yes | yes |
| (1,2) | (1,1) | 3,3 | yes | no |
| (2,0) | (2,2) | 2,2 | yes | yes |
| (2,1) | (1,1) | 3,3 | yes | no |
| (2,2) | (0,0) | 4,4 | yes | no |

SO joint counts: {(0,0):1, (1,1):2, (2,2):3} (note H=K throughout, as forced). DC joint counts: {(4,0):1, (3,1):2, (2,2):3} — and indeed N_DC(4-K, K) for K=0,1,2 gives 1,2,3 = N_SO(K), with orbit maximum D = 4, illustrating 2.2. **Coefficient interpretation:** in the SO enumerator, the coefficient of t^K (equivalently of u^K z^K) counts self-orthogonal assignments of code dimension K; in the DC enumerator, the coefficient of u^K z^H counts dual-containing assignments with code dimension K and hull (= dual) dimension H; at u=z=1 both collapse to the C4 total 6 = tr(B_2^2) for this orbit.

## 2.4 Verdict, genuine content, one notation flag

**R2 is CORRECT** — matrices derived from the criteria with correct orientation, exponents, closing-edge boundary conditions (including L=1), and the symmetry claim proved. Genuine content: C4 states criteria + two total counts + their equality; R2 adds the full dimension distributions inside both families and the distribution-level symmetry. It is a ∗refinement by the same closed-walk mechanism∗, not a new method — that is the honest scope. **Notation flag (not an error):** P1-20 uses the letter D both for the global maximum dimension and (as D(K,H)) for the DC counting

function, in a context where D(C) is also the dual code. If R2 is ever adopted, rename the counting function (e.g. N_DC) as done in 2.2 above. No text is changed now.

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# 3. R3 — the LCD-conditioned code-dimension law

## 3.1 Proof from the exact LCD criterion

The LCD criterion (C3, manuscript Corollary 7.3, proof audited): Hull(a) = 0 iff **every** orbit restriction is constant: alpha = (0,...,0) or alpha = (P,...,P). (Boundary audit of that criterion: for L=1 it reads min(P-a,a)=0 iff a in {0,P}, true for all P; for L >= 2 the propagation argument excludes any interior value and any mixture of 0/P values, for all P >= 1. No parity condition appears anywhere.)

Therefore the LCD event in the full assignment space is the Cartesian product over orbits of the 2-element sets {0-orbit, P-orbit}. The uniform distribution on a Cartesian product, conditioned on a Cartesian-product event, is the product of the uniform distributions on the factors. Hence, conditionally on LCD:

- the orbit restrictions are **independent**;
- each orbit takes its two admissible tuples with probability 1/2 each;
- the code contribution of the orbit is P w_O L_O (for alpha = 0: w ∗ sum_i P) or 0 (for alpha = P), each with probability 1/2.

That is, with independent xi_O ~ Bernoulli(1/2), dim C | LCD =d sum_O (P w_O L_O) ∗ xi_O. The probability generating identity E[u^{dim C}|LCD] = 2^{-#orbits} prod_O (1 + u^{P w_O L_O}) is immediate, and:

```
E[dim C | LCD]  = (1/2) sum_O P w_O L_O = D/2,
Var(dim C | LCD) = sum_O (P w_O L_O)^2 * (1/4) = (P^2/4) * sum_O (w_O L_O)^2,
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using independence (variances add) and Var(xi_O) = 1/4. ∎

## 3.2 P odd? Boundary cases

**P may be odd — no restriction.** The proof uses only P >= 1 integer; parity never enters (contrast the L=1 hull ∗variance∗ in C6, which is parity-split — a different quantity). In the code model P = p^rho may be an odd prime power (e.g. P=3,9) or a power of 2; the combinatorial statement is identical. Boundary cases: P=1 (both binary tuples are LCD since min(1-a,a)=0; contributions wL or 0); L=1 (two point masses at Pw and 0); arbitrary w_O >= 1 (pure scaling of the Bernoulli weights); zero orbits (empty sum: dim C = 0, D = 0, mean/variance 0 — formulas agree, degenerate but consistent). All were included in the exact checks (Section 6), including odd P=5 with mixed weights: configuration (L,w) = (1,2),(2,1) gives LCD count 4, conditional mean 10, variance 50, matching (P/2)(2+2) = 10 and (25/4)(4+4) = 50.

## 3.3 Verdict and duplication assessment

**R3 is CORRECT.** Duplication assessment (the important part): R3 is ∗exactly∗ N3 normalized by the C3 count, plus its first two moments. The Bernoulli representation and the mean/variance are one-paragraph consequences of C3 + uniformity. So: relative to **C3**, R3 adds real distributional content (a full conditional law, not just the count 2^{#orbits}); relative to **N3**, it adds only the normalization and moments. **R3 and N3 must not be adopted as two separate results** — the adoptable item is "N3 in conditional-law form (R3)". Classification: correct, useful, low-depth corollary of the frozen C3.

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# 4. R4 — the orbit-length recurrence

## 4.1 Derivation with dimensions and signs made explicit

Fix P, w. T = T^{(P)}_w(u,z) is an m x m matrix with **m = P+1**, over the commutative ring R = Z[u,z]. Its characteristic polynomial chi_T(lambda) = det(lambda I - T) is monic of degree m with

coefficients in R. Write it as chi_T(lambda) = lambda^m - c_1 lambda^{m-1} - ... - c_m (so c_j = -[lambda^{m-j}] chi_T). Cayley–Hamilton over a commutative ring gives chi_T(T) = 0, i.e. T^m = c_1 T^{m-1} + ... + c_m I. Multiplying by T^L (L >= 0) and taking the trace, with s_L := tr(T^L) for L >= 1 and the **formal** s_0 := tr(I) = m = P+1:

```
s_{L+P+1} = c_1 s_{L+P} + ... + c_{P+1} s_L        for all L >= 0.   ∎
```

The coefficients c_j depend on (P, w, u, z) (through T_1(u^w, z^w) for general w) but **not** on L. "Order at most P+1" is the correct wording: the trace sequence may satisfy a lower-order recurrence for special (u,z); minimality is not claimed and not needed. Note L=0 is not an orbit length; s_0 is only the auxiliary initial value that makes the recurrence valid from L=0.

### 4.2 Independent verification at P=1

For w=1, T = [[u, 1], [u z, 1]] (Proposition 6.4). Direct computation: det(lambda I - T) = (lambda - u)(lambda - 1) - u z = lambda^2 - (1+u) lambda + u(1 - z). In the sign convention of 4.1, c_1 = 1+u, c_2 = -u(1-z), so the predicted recurrence is s_{L+2} = (1+u) s_{L+1} - u(1-z) s_L. Initial values: s_0 = 2 (trace of I_2), s_1 = tr(T) = 1 + u. Independent check at the next step: T^2 = [[u^2 + u z, (1+u)], [u z (1+u), 1 + u z]]-diagonal entries give tr(T^2) = 1 + u^2 + 2 u z, while the recurrence predicts (1+u)(1+u) - 2u(1-z) = 1 + u^2 + 2 u z. Agreement. (The full two-variable polynomial recurrence was also checked for L = 0..6 by the P1-20 validator and again in Section 6.) The closed form s_L = lambda_+^L + lambda_-^L with lambda_{+/-} = ((1+u) +/- sqrt((1-u)^2 + 4 u z))/2 holds in the quadratic extension R[sqrt((1-u)^2+4uz)]: both sides satisfy the same recurrence with the same initial values 2 and lambda_+ + lambda_- = 1+u; this argument is valid also when lambda_+ = lambda_- (discriminant 0), where it reduces to s_L = 2 lambda^L. General w: substitute (u,z) -> (u^w, z^w). ∎

### 4.3 Generic existence vs. new closed form — the honest separation

Cayley–Hamilton guarantees a linear recurrence with polynomial coefficients for the trace sequence of **any** finite matrix; that part of R4 is generic linear algebra, and for P >= 2 the coefficients c_j are ∗not∗ given in closed form in P1-20 (they are computable, e.g. from s_1,...,s_{P+1} via Newton identities / Faddeev–LeVerrier, but computing them is not a closed form). The concrete new content of R4 is therefore exactly: (i) the observation that the orbit enumerators at fixed P form a linear-recurrent sequence in L of order <= P+1 (nothing in C1–C7/N1–N3 relates different orbit lengths), useful for long-orbit computation; and (ii) the explicit two-term recurrence and closed form at P=1. **R4 must not be presented as a general closed form for tr(T^L).** With that scope, **R4 is CORRECT**.

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## 5. Comparison with C1–C7 and N1–N3; what is genuinely new

**Proposed | vs. existing statements | Genuinely new content (audited)**

R1 | C2 gives the code marginal product; nothing joint at fixed H | Joint coefficient symmetry N(K,H)=N(D-K,H); conditional hull law equal at K and D-K (R1.1). **New.**

R2 | C4 = criteria + counts at u=z=1 + equality of totals | Full dimension distributions inside SO/DC families; distribution-level symmetry N_SO(K)=N_DC(D-K,K). **New as distributions; same closed-walk mechanism as C4.**

R3 | C3 = count only; N3 = unnormalized product | As a conditional ∗law with mean/variance∗: new vs. C3; **vs. N3 it is a restatement in normalized form** — adopt at most one of R3/N3.

R4 | T5 evaluates one L; no cross-L statement exists | Recurrence across orbit lengths (generic CH) + explicit P=1 closed form. **New but low-depth; no general closed form.**

(P1-20 Section 5) | N1 (Cov=0) | The exact coefficient formula for `E[H | K] is definitional; the ∗counterexample∗ (conditional means 0,1,1,0 at P=1,L=3) is the valuable part: it refutes the constant-conditional-mean strengthening and fences N1. Consistent with this audit (R1.1 explains the symmetry 0,1,1,0` of that pattern).

(P1-20 Section 6) | C5, C6, N2 | Var(dim C +/- dim Hull) = Var C + Var H and constant linear predictor are one-line consequences of N1+N2+C6. Correct; summary value only.

C7 (EAQECC) | — | R1–R4 have **no** bearing on C7; distances remain external. Nothing in P1-20 was claimed there, correctly.

No item among R1–R4 is a ∗silent∗ restatement of a frozen corollary: R2 at u=z=1 and R3 at u=1

collapse to C4/C3 by design (recorded as specializations, not as content), and R3-vs-N3 duplication is between two ∗proposed∗ items, flagged in 3.3.

## 6. New exact checks (corroboration only; fresh parameters, audit-side code)

Fresh enumeration code written for this audit (not imported from validate_p1_20.py), exact integer/Fraction arithmetic; P=6 instances test the polynomial identity only (not a prime power — same status as the manuscript's T5 remark):

- **R1 + R2 + R3 (orbit level)** at (P,L,w) = (5,3,3), (5,5,1), (6,4,2), (7,3,3), (4,5,2): R1 symmetry holds coefficientwise in every case; at (5,3,3), (6,4,2), (7,3,3) both R2 restricted traces equal brute-force restricted enumeration exactly; per-orbit LCD admissible code contributions are exactly {0, P w L} in every case. All checks passed.
- **R3 at odd P=5**, configuration (L,w)=(1,2),(2,1) by complete enumeration: LCD count 4, conditional mean 10, variance 50 — equal to the formulas (3.2).
- **R4 evaluated recurrence**: at (u,z) = (2,3), the trace sequences for P=2 (s_0..s_9 = 3, 11, 185, 2099, ...) and P=3 (4, 27, 1673, 52731, ...) satisfy the order-(P+1) recurrence built from the evaluated characteristic polynomials; the P=1 polynomial recurrence was rechecked for L=0..6 (s_1 = 1+u, s_2 = 1+u^2+2uz). All checks passed.
- The committed P1-20 validator was also re-run unmodified: ALL P1-20 VALIDATION CHECKS PASSED (173 checks). Finite checks corroborate; the proofs in Sections 1–4 are the evidence.

**No counterexample to any of R1–R4 was found**, in the proofs or on the grids. Accordingly there is no correction proposal for any statement; the only recommended changes are the wording fences in Section 7, and they concern future adoption text, not the P1-20 report's correctness.

## 7. Verdicts and wording fences (for any future adoption decision — none made here)

**Result | Verdict | Required scope wording if ever adopted**

R1 (+R1.1) | **Correct; proof complete; genuinely new joint content.** Strongest of the four as a statement. | State D = n sum_s m_s; keep "joint/fixed-hull" emphasis vs. C2.

R2 (+symmetry) | **Correct; matrices/orientation/boundaries verified; genuine distribution refinement of C4.** | Rename the DC counting function (N_DC) to avoid the D overload; present as refinement of C4, u=z=1 specialization noted.

R3 | **Correct for all P>=1 (odd included); low-depth corollary of C3.** | Adopt only as "N3, conditional-law form" — never as a second result alongside N3.

R4 | **Correct; generic (Cayley–Hamilton) + explicit only at P=1.** | Say "order at most P+1"; never "closed form" for P>=2; s_0 flagged as formal.

Rejected / not promoted (unchanged from P1-20, re-confirmed here): constant conditional hull mean at fixed code dimension (**false**, counterexample in P1-20 Section 5, consistent with R1.1's symmetry); independence of the two dimensions (**false**); palindromy in the hull variable (**false**, {0:2, 1:6} at P=1,L=3); naive monotonicity in L (ill-posed); Perron–Frobenius growth (standard background, not a result).

**Overall audit conclusion:** R1–R4 are mathematically correct as stated in P1-20, under exactly the hypotheses recorded above. Their depths differ: R1 is a clean new structural symmetry; R2 is a genuine distributional refinement of C4 by a known mechanism; R3 is N3 in its useful normalized form; R4 is a generic recurrence with an explicit simple-root case. All remain **PROPOSED**; the frozen blueprint and manuscript are untouched by this audit, and no novelty or publication-readiness claim is made (Gate G1 DEFERRED).